

PROGRAMMING FOR PROBLEM SOLVING

UNIT 1 C Fundamentals & Decision Making

PREMANAND S

Assistant Professor

Department of Computational Intelligence

SRM Institute of Science and Technology

Kattankulathur Campus

First Year Engineering

Problem 1 – EASY: Menu Selection

Scenario:

Create a simple college canteen menu.

The user enters a choice:

- **1** → Tea
- **2** → Coffee
- **3** → Juice
- **4** → Sandwich

Display the selected item.

For any other choice, display **Invalid Choice**.

Task:

Write a C program using the `switch` statement.

Problem 2 – Calculator

Scenario:

Build a simple calculator.

The user enters:

- Two numbers
- An operator: +, -, *, /

The program should perform the selected operation.

Example:

For division, if the second number is 0, display:

Cannot divide by zero

For any other operator, display:

Invalid Operator

Task:

Use the `switch` statement to select the operation.

Problem 3 – HOTS: ATM Transaction Menu

Scenario: Design a simple ATM transaction menu.

ATM MENU

1. Check Balance
2. Deposit
3. Withdraw
4. Exit

The user selects an option.

Rules:

- **1 – Check Balance:** Display the current balance.
- **2 – Deposit:** Ask for the deposit amount and add it to the balance.
- **3 – Withdraw:**
 - Check whether the amount is positive.
 - Check whether sufficient balance is available.
 - If sufficient balance exists, perform the withdrawal.
 - Otherwise display **Insufficient Balance**.
- **4 – Exit:** Display **Thank You!**

Problem 1: Library Entry

Scenario

A college library allows a student to enter the **Exam Preparation Section** only if:

- The student has a valid college ID.
- If the ID is valid, check whether the student has a library membership.
- Entry is allowed only when both conditions are satisfied.

Task

Write a C program using a **nested** if statement to display:

“Entry Allowed” or **“Entry Not Allowed”**

Problem 2: Online Examination

Scenario

A student wants to attend an online examination.

The system should:

- 1 First check whether the student has **registered for the exam**.
- 2 If registered, check whether the **exam fee is paid**.
- 3 If both conditions are satisfied, display **“Exam Login Allowed”**.
- 4 If the student is not registered, display **“Please Register First”**.
- 5 If registered but the fee is not paid, display **“Pay Exam Fee”**.

Task

Write a C program using a **nested if--else** statement to implement this decision.

Problem 3: Smart ATM

Scenario

An ATM has to process a withdrawal request.

The program receives:

- `pinCorrect`: 1 for correct PIN, 0 otherwise
- `balance`: available account balance
- `amount`: amount requested

The ATM should:

- 1 First check whether the **PIN is correct**.
- 2 If correct, check whether sufficient **balance** is available.
- 3 If sufficient balance is available, check whether the amount is a **multiple of Rs. 100**.
- 4 If all conditions are satisfied, display **“Withdrawal Successful”**.